Valuation balance and local lifting for cyclic n-roots

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Abstract

In its usual complex form, the Björck–Saffari conjecture asks whether the cyclic n-roots are finite in number whenever n is square-free. This is proved for prime n; among composite square-free orders, the sources we read settle only $n = 6$ and $n = 10$, by computer algebra. Haagerup's prime-order proof starts from the fact that an infinite solution set yields disjoint supports in time and frequency, and Zhou showed that such supports exist at every composite order. We show that the Fourier equations force more. Along any Laurent-series solution, the multiset of coordinate valuations equals the multiset of valuations of its Fourier transform. This is a consequence of a residue principle for master functions, and we claim no priority for it. It gives exact complementary supports of equal size at every proper valuation gap. For a one-dimensional endpoint kernel and a nondegenerate projective normal critical point, we reduce local lifting to one analytic function and give its first two scalar conditions, $\kappa = 0$ and $\mu = 0$. At order 26, one fixed support pair admits nondegenerate normal critical points with $\kappa \neq 0$ and others with $\kappa = 0$; an affine reflection forces the latter cancellation exactly. At two such points a rigorous π-adic computation shows $\mu \neq 0$, so no solution branch leaves them in the prescribed directions. Thus κ alone cannot exclude every endpoint, while μ excludes these. We prove no bound on the number of conditions needed in general, and no finiteness theorem for any composite order.

1 The question and the contribution

A cyclic n-root is a vector $a \in \mathbb{C}^n$ satisfying

$$\sum_{j=0}^{n-1} a_j a_{j+1} \cdots a_{j+r-1} = 0 \quad (1 \le r < n), \qquad \prod_{j=0}^{n-1} a_j = 1,$$

with indices modulo n. If m^2 divides n for some $m > 1$, there are infinitely many cyclic n-roots (Backelin [2]; Björck and Saffari for the unimodular solutions [3]). Björck and Saffari conjectured that for square-free n the unimodular solutions with leading entry 1 are finite in number [8, Conj. 1.1]; the form for all complex solutions is the standard restatement [11]. Haagerup proved finiteness, and a count with multiplicity, for prime order [9]. For composite square-free orders, the sources we read establish finiteness only for $n = 6$ and $n = 10$, by Gröbner-basis computation; at $n = 14$ a count of isolated roots is reported, but finiteness is not established there (see the table and its footnote in [8]). The question for $2p$, with p an arbitrary odd prime, motivates this note but is not resolved here.

Put $\zeta = e^{2\pi i/n}$ and $F = (\zeta^{ij})_{0 \le i,j < n}$. We use the unnormalized Fourier matrix, bilinear transpose, and coordinatewise reciprocals. Work where all coordinates of x and $f = Fx$ are nonzero.

The equations are

$$E(x) := F(Fx)^{-1} - x^{-1} = 0. \tag{1}$$

After fixing $x_0 = 1$, these are the Fourier equations for cyclic roots: set $a_j = x_{j+1}/x_j$. Indeed, $F^{-1}(x^{-1}) = (Fx)^{-1}$ says that the cyclic correlations $\sum_j x_{j+r}/x_j$ vanish for $r \neq 0$. This correspondence is credited to the existing literature, particularly [9].

Haagerup's limiting-support argument produces disjoint supports in time and frequency from an infinite solution set. Zhou proves that such pairs exist for every composite order [11]. Our valuation statement retains multiplicities and all the gaps, yielding exact complements of equal size. These conditions are necessary, not sufficient for a branch.

We then express the normal equations at a boundary point as the critical equations for a product of linear forms. Such master functions are standard; see [4]. An implicit-function reduction supplies the concrete scalars κ, μ. Puiseux expansions of cyclic n-roots were computed earlier from initial forms by Adrovic and Verschelde, who test whether a second term exists, starting from a regular isolated root of the initial system [1, Sec. 5.1]; κ and μ are compatibility conditions of this kind, written explicitly for nondegenerate normal critical points at a valuation gap. The order-26 example shows both nonvanishing and symmetry-forced vanishing of the first scalar; at two of the symmetric points a rigorous π-adic computation shows $\mu \neq 0$, so no branch leaves them in the prescribed directions. The reflection used there is, up to a scalar factor, a composite of the shifts, modulations and dilations that Führ and Rzeszotnik list for unimodular solutions [8, Sec. 10.3]; the same formulas act on all solutions. The Fourier symmetries, master-function framework, and implicit-function method are not claimed as new.

Theorem 2.1 is a consequence of a general residue principle: when a formal curve of critical points of a master function $\prod f_i^{\alpha_i}$ tends to boundary divisors E of a smooth simple normal-crossings compactification, the weights satisfy $\sum_i \operatorname{ord}_E(f_i)\alpha_i = 0$ for each of them [10, Prop. 2.10]. Apply it to the $2n$ forms x_i and $(Fx)_k$, with weights $+1$ on $(Fx)_k$ and -1 on x_i (total weight 0, so the master function lives on projective space), on the maximal projective wonderful model of their arrangement [5, §§3–4] ("wonderful model" is De Concini and Procesi's technical name for this blow-up compactification); see also [6, Thm. 2.5, §§4.1–4.2] and, for its fan, [7, Prop. 3.6, Thm. 4.1]. Its boundary divisors are indexed by the nonempty proper flats. Each set $\{\ell : \nu(\ell) > c\}$ is a flat, since a form in the span of forms of order $> c$ has order $> c$, and the solution curve meets its divisor, along which ℓ vanishes to order 1 exactly when ℓ lies in the flat. The residue condition there says that the flat contains as many forms $(Fx)_k$ as forms x_i, which is the equality of the counts at threshold c. We give a direct proof below and claim no priority for Theorem 2.1. For cyclic n-roots, this note adds the exact supports at every gap, the scalars κ, μ, and the order-26 example; a bounded search, recorded with the note, found none of them stated for cyclic n-roots.

2 Valuations and every gap

Write $\nu(h)$ for the order of a nonzero Laurent series at $t = 0$. The next result uses symmetry and invertibility rather than Fourier entries.

Theorem 2.1 (Valuation multiset)**.** *Let F be an invertible symmetric constant matrix over $\mathbb{C}$. Suppose $x \in \mathbb{C}((t))^n$ satisfies (1) and all coordinates of x and Fx are nonzero. Then*

$$\{\nu(x_i) : 0 \le i < n\} = \{\nu((Fx)_k) : 0 \le k < n\}$$

as multisets. This also holds for Puiseux series after a finite change of parameter.

Proof. Let $D = t\,d/dt$, $y = x^{-1}$, and $f = Fx$. Since $F^{-1}y = f^{-1}$, for every integer $r \geq 1$ we have

$$\sum_k \frac{D^r f_k}{f_k} = (FD^r x)^T F^{-1} y = (D^r x)^T y = \sum_i \frac{D^r x_i}{x_i}.$$

If $h = ct^a(1 + O(t))$, then $D^r h/h = a^r + O(t)$. Constant terms give equal power sums of the valuation multisets. Newton's identities for the first n sums determine a multiset of n numbers. A common denominator of Puiseux exponents reduces that case to Laurent series. □

Corollary 2.2 (Exact complementarity at every gap)**.** *Under Theorem 2.1, choose a real c which is not a coordinate valuation, and put*

$$I = \{i : \nu(x_i) < c\}, \qquad J = \{k : \nu((Fx)_k) < c\}.$$

If the cut is proper, then $0 < |I| = |J| < n$, and constant vectors u, v exist with

$$\operatorname{supp} u = I, \quad \operatorname{supp}(Fu) = J, \qquad \operatorname{supp} v = I^c, \quad \operatorname{supp}(F^{-1}v) = J^c.$$

Proof. The cardinalities follow from the multiset theorem. Each coefficient vector $x^{[r]}$ for $r < c$ is supported in I with transform supported in J. There are finitely many such nonzero vectors. Every coordinate of I, and every transform coordinate of J, occurs in at least one. A generic constant linear combination has exactly those supports, since bad combinations form a finite union of proper hyperplanes. Apply the same argument to $y = x^{-1}$ and $F^{-1}y = f^{-1}$ at orders less than $-c$ to obtain the complements. □

Remark 2.3. An infinite normalized cyclic-root variety contains an algebraic curve. On its smooth projective model a nonconstant coordinate has a pole. The Laurent expansion there, with $x_0 = 1$, has a nonconstant valuation vector and a proper gap. Thus the corollary gives necessary data for an infinite solution set, without asserting that arbitrary such data occur along a solution curve.

3 The normal equations and two scalar conditions

Continue with any invertible symmetric F. Fix $|I| = |J| = m$, put $q = n - m$, and suppose

$$\operatorname{supp} u = I, \quad \operatorname{supp} f = J, \quad f = Fu, \quad A = F[J^c, I], \quad \operatorname{rank} A = m - 1. \tag{2}$$

The right kernel is the line u_I. Let the columns of $G \in \mathbb{C}^{q \times d}$ span $\ker A^T$, where $d = q - m + 1$. Set

$$V = F[I^c, J^c]G, \quad g_{J^c} = Gz, \quad g_J = 0, \qquad v = Fg.$$

Work where all entries of Gz and $Vz = v_{I^c}$ are nonzero. Define

$$\Psi(z) = \sum_{k \in J^c} \log(Gz)_k - \sum_{i \in I^c} \log(Vz)_i. \tag{3}$$

Only logarithmic derivatives are used. The equal factor counts make this a projective critical-point problem.

Proposition 3.1. *A vector w with*

$$w_{I^c} = v_{I^c}^{-1}, \qquad (Fw)_{J^c} = g_{J^c}^{-1} \tag{4}$$

exists exactly when $\nabla\Psi(z) = 0$, and is unique modulo u. Set

$$H = V^T \operatorname{diag}(v_{I^c}^{-2})V - G^T \operatorname{diag}(g_{J^c}^{-2})G, \qquad L = \operatorname{diag}(v^2) - F \operatorname{diag}(g^2)F.$$

At a critical point, $\dim\ker L = 1 + \dim\ker H$. *If* $\operatorname{rank} H = d - 1$, *then* $\ker L = \operatorname{span}\{u, w\}$.

Proof. Solving for w_I is an A-system with complete left-kernel conditions $G^T g^{-1} - V^T v^{-1} = 0$, precisely $\nabla\Psi = 0$. Let $Nr = (r_{I^c}, (Fr)_{J^c})$ and $D = \operatorname{diag}(v_{I^c}^2, -g_{J^c}^2)$. Then $L = N^T D N$, $\ker N = \mathbb{C}u$. The columns of $\Lambda = (V^T, -G^T)^T$ span $\ker N^T$; their number is $2q - (n-1) = d$. Also $\Lambda^T D^{-1} \Lambda = H$. Indeed $\dim\ker N^T = 2q - \operatorname{rank} N = 2q - (n-1) = d$, and the d columns of Λ are independent because those of G are. Now $Lr = 0$ means $DNr = \Lambda b$. Such r exists exactly when $Hb = 0$, and is unique modulo u. Finally $Lu = 0$, and (4) gives $DNw = (v_{I^c}, -g_{J^c}) = \Lambda z$, so $Lw = N^T \Lambda z = 0$; u, w are independent since w is nonzero on I^c. □

Fix a critical point z of Ψ and assume projective nondegeneracy: $\operatorname{rank} H = d - 1$. Put

$$h = F(1/f \text{ on } J) - (1/u \text{ on } I), \qquad \kappa = w^T h, \tag{5}$$

with zero entries elsewhere. Since $u^T h = |J| - |I| = 0$, κ is independent of the choice of w. Symmetry of L gives

$$Ls = -h \text{ is solvable} \quad \Longleftrightarrow \quad \kappa = 0. \tag{6}$$

When this holds, define

$$\mu = \sum_{j \in J} \left(\frac{(Fw)_j}{f_j} \right)^2 - \sum_{i \in I} \left(\frac{w_i}{u_i} \right)^2 + s^T L s. \tag{7}$$

Changing s by $\ker L$ has no effect. Replacing w by $w + cu$ changes the two sums by $2c\kappa + c^2(|J| - |I|) = 0$. Under $u \mapsto au$, $g \mapsto bg$, we have $\kappa \mapsto (ab)^{-1}\kappa$ and, when defined, $\mu \mapsto (ab)^{-2}\mu$. Their vanishing is independent of scale.

Theorem 3.2 (Local scalar reduction)**.** *At such a critical point, under (2), the nonzero-coordinate conditions, and projective nondegeneracy, there is a convergent effective function $Q_{\text{eff}}(\tau)$ with $Q'_{\text{eff}}(0) = \kappa$. If $\kappa = 0$, then*

$$Q'_{\text{eff}}(\tau) = -\mu\tau + O(\tau^2).$$

A formal branch with boundary direction u and first transverse direction w exists exactly when $Q'_{\text{eff}} \equiv 0$. Here such a branch is a solution $x(t)$ of (1) over $\mathbb{C}((t))$, with all coordinates of x and Fx nonzero, which after multiplication by a scalar series satisfies $\ell_u(x) = 1$ and $x = u + \tau(t)w + \xi(t)$, where ℓ_u, ℓ_w, T are as in the proof, $\xi \in T[[t]]$, $\tau \in t\mathbb{C}[[t]]$ is nonzero, and $\nu(\xi) > \nu(\tau)$.

Proof. Choose linear forms ℓ_u, ℓ_w dual to u, w on their span and put $T = \ker \ell_u \cap \ker \ell_w$. In the slice $\ell_u(x) = 1$, write $x = u + \tau(w + \eta)$, $\eta \in T$. The rational function $\mathcal{R}(x) = \prod_k (Fx)_k / \prod_i x_i$ has logarithmic gradient $E(x)$. Cancelling its τ^q factors gives, up to a constant, the logarithm

$$\begin{aligned} Q(\tau, \eta) &= Q_N(\eta) + Q_R(\tau, \eta), \\ Q_N &= \sum_{j \in J^c} \log\left(1 + \frac{(F\eta)_j}{(Fw)_j}\right) - \sum_{i \in I^c} \log\left(1 + \frac{\eta_i}{w_i}\right), \\ Q_R &= \sum_{j \in J} \log\left(1 + \tau \frac{(Fw + F\eta)_j}{f_j}\right) - \sum_{i \in I} \log\left(1 + \tau \frac{w_i + \eta_i}{u_i}\right). \end{aligned}$$

At $(0, 0)$ the η-gradient is zero, by the normal system (4), and its Hessian is $L|_T$. A vector in its radical pairs to zero with T and with u, w, so lies in $\ker L \cap T = 0$. The formal and analytic implicit-function theorems give the same unique $\eta(\tau) \in \tau T[[\tau]]$ with $\partial_\eta Q = 0$. Define $Q_{\text{eff}}(\tau) = Q(\tau, \eta(\tau))$.

For $\tau \neq 0$ these coordinates parametrize the slice near the specified direction. Homogeneity of $\mathcal{R}$ makes criticality on the slice equivalent to $E(x) = 0$. Given a branch as in the statement, put $\eta_b = \xi/\tau \in tT[[t]]$, so that $x = u + \tau(w + \eta_b)$ and both partial derivatives of Q vanish at

$(\tau(t), \eta_b(t))$. Write $\partial_\eta Q(\tau,\eta) = L|_T\eta + \beta(\tau) + O(\tau\eta) + O(\eta^2)$ with $\beta(0) = 0$. Since $\tau(t) \in t\mathbb{C}[[t]]$ and $L|_T$ is invertible, the coefficient of t^k in $\partial_\eta Q(\tau(t), \eta(t)) = 0$ determines the t^k coefficient of η from the lower ones; so this equation has a unique solution in $tT[[t]]$, and $\eta_b = \eta \circ \tau$. Then $Q'_{\mathrm{eff}}(\tau(t)) = \partial_\tau Q + \partial_\eta Q \cdot \eta' = 0$, and a nonconstant $\tau(t)$ cannot annihilate a nonzero power series in τ. A branch therefore implies $Q'_{\mathrm{eff}} \equiv 0$. Conversely that identity supplies the convergent branch $u + \tau(w + \eta(\tau))$, with all required coordinates nonzero for small nonzero τ.

Differentiation gives $Q_\tau(0,0) = w^T h = \kappa$. If $\kappa = 0$, let $s = \eta'(0)$. The transverse equation gives $(Ls + h)|_T = 0$; it also vanishes on u, w, so $Ls = -h$. Finally,

$$Q''_{\mathrm{eff}}(0) = Q_{\tau\tau} + 2h^T s + s^T L s = -\left\{ \sum_J ((Fw)_j/f_j)^2 - \sum_I (w_i/u_i)^2 + s^T L s \right\} = -\mu.$$

This does not restrict intermediate powers in a Laurent parameter. □

4 One order-26 support with opposite outcomes for κ

Set $\omega = e^{2\pi i/13}$, $\zeta = -\omega^7$, and $K = \mathbb{Q}(\omega)$. Take

$$\begin{aligned} I &= \{0,1,2,3,5,7,13,15,17,18,19,20\}, \\ J &= \{3,9,13,14,15,16,18,20,22,23,24,25\}, \\ B = J^c &= \{0,1,2,4,5,6,7,8,10,11,12,17,19,21\}. \end{aligned}$$

For $E = \{0,1,2,3,4,5,6\}$ and $O = \{2,3,4,7,9,10,12\}$, write

$$U(X) = \prod_{e\in E}(X - \omega^e) = \sum a_i X^i, \qquad W(X) = \prod_{o\in O}(X + \omega^o) = \sum b_i X^i.$$

Let u be the length-26 coefficient vector of

$$b_1 U(X) + a_1 W(X) + X^{13}\{b_1 U(X) - a_1 W(X)\}.$$

Exact arithmetic gives $\operatorname{supp} u = I$ and $\operatorname{supp} Fu = J$. Put $D_0 = I \setminus \{20\}$, $T_0 = \{17,19,21\}$, $C_0 = B \setminus T_0$. The normal system uses

$$M = F[D_0, C_0], \qquad G_{T_0} = \mathrm{Id}_3, \qquad G_{C_0} = -M^{-1}F[D_0, T_0]. \tag{8}$$

Under $\zeta \mapsto 5215$ modulo 10037, $\det M = 3759$. Thus $\operatorname{rank} F[B, I] = 11$, and $z = (g_{17}, g_{19}, g_{21})$ are free coordinates. The modular minor proves nonvanishing only; exact cyclotomic identities prove the support zeros.

Theorem 4.1. *These supports and this u admit algebraic nondegenerate projective critical points of (3) with $\kappa \neq 0$. They also admit at least two distinct algebraic nondegenerate projective critical points with $\kappa = 0$. At two of the latter, $\mu \neq 0$; hence no formal branch with boundary direction u and first transverse direction w, in the sense of Theorem 3.2, leaves either of them.*

4.1 A point with $\kappa \neq 0$

Reduce at the split prime 53 with $\zeta \mapsto 4$, $\omega \mapsto 16$. At $z = (1, 8, 19)$, exact finite-field arithmetic gives

$$\nabla\Psi(z) = 0, \qquad H = \begin{pmatrix} 6 & 47 & 5 \\ 47 & 3 & 13 \\ 5 & 13 & 11 \end{pmatrix}, \qquad \kappa = 4.$$

All 28 denominators are nonzero, $\det M = 46$, and the Hessian determinant on $z_0 = 1$ is 23. Hensel's lemma gives a normal critical point over $\mathbb{Q}_{53}$ at which κ is a unit. The cleared two-equation system has an invertible Jacobian, so the point is isolated and algebraic over K. Embed a number field of definition into $\mathbb{C}$ by an embedding extending $\omega \mapsto e^{2\pi i/13}$. This gives the required point, preserving the nonzero coordinates, the Hessian rank and $\kappa \neq 0$. By (13) below, $\kappa(u, Rz) = -\kappa(u, z) \neq 0$. Since κ scales as λ^{-1} under $z \mapsto \lambda z$, Rz could represent the same projective point only if $Rz = -z$, which forces $z_1 = 0$; but z_1 reduces to $8 \neq 0$ modulo 53. So Rz is a second such point.

4.2 A symmetry that makes κ odd

Define monomial involutions, with indices modulo 26, by

$$(\mathcal{U}x)_i = \zeta^{-10-12i} x_{20-i}, \qquad (\mathcal{S}g)_k = \zeta^{-10-20k} g_{12-k}. \tag{9}$$

Direct multiplication gives

$$\mathcal{U}^2 = \mathcal{S}^2 = 1, \qquad F\mathcal{U} = \mathcal{S}^{-T} F, \qquad F\mathcal{S} = \mathcal{U}^{-T} F. \tag{10}$$

The supports are preserved since $I = 20 - I$ and $J = 12 - J$. Substitution in the polynomial for u, in $\mathbb{Z}[\omega]$, gives

$$\mathcal{U}u = -u. \tag{11}$$

Equivalently, $(\mathcal{U}_0 x)_i = \zeta^{-12i} x_{20-i}$ acts on u by $-\zeta^{10}$. On the free coordinates, $\mathcal{S}$ induces

$$R(z_0, z_1, z_2) = (\zeta^{14} z_2, z_1, \zeta^{12} z_0). \tag{12}$$

The products in (3) are permuted and multiplied by constants, so R preserves the critical locus. If (u, g, w) satisfies (4), then $(\mathcal{U}u, \mathcal{S}g, \mathcal{U}w)$ does also. Moreover $h(\mathcal{U}u) = \mathcal{U}^{-T} h(u)$, by (10). Therefore $\kappa(\mathcal{U}u, \mathcal{S}g) = \kappa(u, g)$. Equation (11) and scaling in u imply

$$\kappa(u, Rz) = -\kappa(u, z) \tag{13}$$

on the normal critical locus. An exactly R-fixed representative has $\kappa = 0$. Projective fixedness without control of scale would not alone justify that conclusion.

4.3 Two fixed Hensel lifts

Let $\pi = 1 - \omega$, with $\nu_\pi(\pi) = 1$. The basis (8) is integral at π, with the free rows the identity. Divide each row of G, V by its largest common power of π and reduce. Up to nonzero row constants, the G rows, in increasing B order, are

$$\begin{gathered}(1,4,7),(1,1,6),(1,3,9),(1,0,0),(1,7,4),(0,1,0),\\ (1,5,10),(0,0,1),(1,9,3),(1,11,11),(1,8,2),\\ (1,0,0),(0,1,0),(0,0,1).\end{gathered} \tag{14}$$

All G rows have valuation zero. Every normalized V row is proportional to $\ell = (1, 6, 1)$; the row valuation is 4 at the nine even indices of I^c and 8 at its five odd indices. The accompanying certificate program verifies these finite cyclotomic arithmetic statements. In particular, write each entry as $\sum_{j=0}^{11} c_j \pi^j$, $c_j \in \mathbb{Q}$. Its valuation is $\min_{c_j \neq 0}(12 v_{13}(c_j) + j)$: the term valuations are distinct modulo 12, so the minimum cannot cancel. Initial units follow from $13/\pi^{12} \equiv -1 \pmod{\pi}$. Thus

these reductions can be checked in the exact number field, without a precision cutoff. They give the reduced logarithmic derivative of

$$\overline{\Psi} = \sum_{r \text{ in } (14)} \log(rz) - 14\log(z_0 + 6z_1 + z_2).$$

In $\mathbb{F}_{169} = \mathbb{F}_{13}[t]/(t^2 - 2)$ take

$$\bar{z}_\pm = (9 \pm 4t,\ 8 \pm 3t,\ 9 \pm 4t).$$

They lie on $\ell z = 1$, all 28 normalized forms are nonzero, and the gradient vanishes. On $z_2 = 1 - z_0 - 6z_1$ the Hessian is

$$\begin{pmatrix} 7 \pm 9t & 8 \pm t \\ 8 \pm t & 4 \end{pmatrix}, \qquad \det = 1 \pm 7t \neq 0,$$

whose norm to $\mathbb{F}_{13}$ is 7.

For the lift use the exactly R-invariant affine chart

$$L(z) = z_0 + 6z_1 + \zeta^{14} z_2 = 1.$$

This reduces to $\ell z = 1$. On each residue disc the normalized denominators are units. Hensel's lemma therefore gives a unique critical lift on each disc in the unramified quadratic extension of K_π. The reduction of R swaps z_0, z_2 and fixes both residual points. Since R preserves $L = 1$, the equations, and each disc, uniqueness forces both lifts to be exactly R-fixed. Equation (13) proves $\kappa = 0$ exactly. As above, nonsingularity of the polynomial system makes both points algebraic over K. Embedding a common number field of definition into $\mathbb{C}$ in the same way gives two distinct complex endpoints.

4.4 The second scalar at the fixed lifts

At each of the two lifts, $\kappa = 0$, so μ is defined by (7). Normalize u as above, $g = Gz$ in the chart $L(z) = 1$, and $w_{20} = 0$ (allowed since $u_{20} \neq 0$); solve $Ls = -h$ with $s_4 = s_{20} = 0$. This is possible because $\ker L = \operatorname{span}\{u, w\}$ and $u_4 = 0$ while $u_{20} w_4 \neq 0$, so evaluation at the coordinates $4, 20$ is injective on $\ker L$. The accompanying program `certify_mu.py` computes in $\mathbb{Z}_{13}[\pi, t]$ with $t^2 = 2$ modulo π^{240}, carrying for every quantity a valuation and a guaranteed absolute precision (conservative π-adic ball arithmetic, no floating point). Newton's method in the chart, started at $\bar{z}_\pm$, reaches gradient valuation 232; since the Hessian is a unit on the whole residue disc, the exact Hensel lift lies within $O(\pi^{232})$ of the iterate, and all later quantities are computed on that enclosure. It finds

$$v_\pi(\mu_\pm) = -14, \qquad \pi^{14}\mu_\pm \equiv 11 \pm 8t \pmod{\pi},$$

with absolute precision 193; the residues have norm $6 \neq 0$ in $\mathbb{F}_{13}$. Thus $\mu \neq 0$ at both lifts. Being nonzero is independent of the normalizations, by the scaling law after (7), and is preserved by the embedding into $\mathbb{C}$, since μ is an algebraic number. By Theorem 3.2, $Q'_{\text{eff}}(\tau) = -\mu\tau + O(\tau^2)$ is not identically zero, so no branch with the directions u, w exists. This proves Theorem 4.1. It excludes neither other directions through these boundary points nor other strata.

5 What remains open

The multiset theorem supplies a necessary condition at every gap, without classifying which support pairs can come from a single Laurent solution. The local reduction is exact under its rank and

nondegeneracy hypotheses. Degenerate normal critical points, larger endpoint kernels, and branches with more than two valuation levels require separate treatment.

Even in the nondegenerate case, $Q'_{\mathrm{eff}} \equiv 0$ gives no bound on how many coefficients must be examined. The order-26 symmetry proves that testing κ alone cannot exclude every endpoint; at the two certified points μ does exclude the prescribed branches. Whether some fixed number of coefficients always suffices is open. A uniform finiteness argument needs an additional theorem forcing some coefficient to be nonzero, with control over the endpoints and orders covered. The present results do not supply that theorem.

Computational evidence and authorship. The certificates use exact cyclotomic arithmetic, finite fields and conservative π-adic ball arithmetic. A finite-precision zero is not used to prove $\kappa = 0$; symmetry and Hensel uniqueness prove that identity. Two AI systems, ChatGPT/Codex (OpenAI) and Claude (Anthropic), worked on this program in parallel under the author's direction and contributed to finding the results, attacking and checking the proofs, verifying the certificates, searching prior work and drafting the text. AI is not an author or a referee, AI review is not peer review, and agreement between AI systems is not independent validation. The author is responsible for the claims and their presentation.

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